

Power equations



1 Solve the following equations:

a $x^5 = 3^5$

b $8^{-7} = x^{-7}$

c $x^3 = \left(\frac{8}{5}\right)^3$

d $x^{-7} = \frac{1}{6^7}$

e $3(x^{-9}) = \frac{3}{2^9}$

f $x^{\frac{1}{3}} = \sqrt[3]{6}$

g $\sqrt[3]{5} = x^{\frac{1}{3}}$

h $\frac{1}{2^5} = x^{-5}$

Exponential equations

2 Solve the following exponential equations:

a $4^x = 4^8$

b $3^x = 3^{\frac{2}{9}}$

c $3^x = 27$

d $7^x = 1$

e $8^x = \frac{1}{8^2}$

f $3^y = \frac{1}{27}$

g $7(4^x) = \frac{7}{4^3}$

h $5^x = \sqrt[3]{5}$

i $30^n = \sqrt[3]{30}$

j $10^x = 0.01$

3 Consider the following equations:

i Rewrite each side of the equation with a base of 2.

ii Solve for x .

a $8^x = 4$

b $16^x = \frac{1}{2}$

c $\frac{1}{1024} = 4^x$

d $(\sqrt{2})^x = \sqrt[5]{32}$

4 Solve the following exponential equations:

a $(\sqrt{6})^y = 36$

b $(\sqrt{2})^k = 0.5$

c $9^y = 27$

d $3^{5x-10} = 1$

e $25^{x+1} = 125^{3x-4}$

f $\frac{1}{3^{x-3}} = \sqrt[3]{9}$

g $\left(\frac{1}{9}\right)^{x+5} = 81$

h $\left(\frac{1}{8}\right)^{x-3} = 16^{4x-3}$

i $\frac{25^y}{5^{4-y}} = \sqrt{125}$

j $8^{x+5} = \frac{1}{32\sqrt{2}}$

k $30 \times 2^{x-6} = 15$

l $2^x \times 2^{x+3} = 32$

m $3^x \times 9^{x-k} = 27$

n $a^{x-1} = a^4$

o $a^{x+1} = a^3\sqrt{a}$

p $3^{x^2-3x} = 81$



Exponential equations and logarithms

5 Consider the following equations:

i Rearrange the equation into the form $x = \frac{\log A}{\log B}$.

ii Evaluate x to three decimal places.

a $13^x = 5$ b $5^x = \frac{1}{11}$ c $3^x = 2$ d $4^x = 6.4$

e $(0.4)^x = 5$ f $5^x + 4 = 3129$ g $2^{-x} = 6$ h $27^x + 4 = 19\,211$

6 For each of the following incorrect sets of work:

i Which step was incorrect? Explain your answer.

ii Rearrange the original equation into the form $a = \frac{\log A}{\log B}$.

iii Evaluate a to three decimal places.

a	$9^a = 40$	b	$2^a = 89$
	$\log 9^a = \log 40$ (1)		$\log 2^a = \log 89$ (1)
	$a + \log 9 = \log 40$ (2)		$a \log 2 = \log 89$ (2)
	$a = \log 40 - \log 9$ (3)		$a = \log_{89} 2$ (3)
	≈ 0.648 (4)		≈ 0.154 (4)

7 Find the x -coordinate of the point of intersection of the graphs of $y = 2^{5x}$ and $y = 4^{x-3}$.

Logarithmic equations

8 Solve for y in each of the following logarithmic equations:

a $\log_7 y = 5$	b $\log_{16} y = \frac{1}{2}$
c $\log_y 8 = 3$	d $2 \log_2 y + 3 \log_2 y = 15$
e $\log_6 y^2 = 4$	f $\log_2 y^4 + \log_2 y = 10$

9 Solve for x in each of the following logarithmic equations:



a $\log_2 16 = x$

c $\log_3 x = 4$

e $\log_{0.5} x = 10$

g $\log_x 64 = 2$

i $\log_9 (7x + 5) = 4$

k $\log_{10} (3x + 982) = 3$

m $11 \log_5 (x - 12) = 33$

o $\log_2 (\log_2 x) = 0$

q $\log_9 (10x - 3) = \log_9 (5x + 4)$

b $\log_{12} 144 = x$

d $2 \log x = 4$

f $\log_{16} x = \frac{1}{4}$

h $\log_x 125 = 3$

j $\log_6 (3x - 9) = 2$

l $6 \log_4 (2x) - 18 = 0$

n $\log_2 \left(\sqrt{2x^3} \right) + 1 = 4.5$

p $\log_2 (3x + 35) = \log_2 5$

10 Solve for x in each of the following logarithmic equations:

a $\log_7 (9x - 2) = \log_7 (8x + 9)$

c $\log_2 x - 4 = \sqrt{\log_2 x - 4}$

e $\log_{10} x + \log_{10} 6 = \log_{10} 48$

g $\log_4 45 - \log_4 9x = 2$

i $\log x + \log (x + 3) = 1$

k $\log_5 (4x^2) - \log_5 x = 3$

m $\log_8 (5x + 12) = \log_8 (x + 6) - \log_8 3$

o $\log_4 (x + 3) + \log_4 (x - 3) = 2$

b $\log_6 x = \sqrt{\log_6 x}$

d $3 \log x = \log 125$

f $\log_7 2x + \log_7 3 = 3$

h $\log_4 x = \log_4 7 - \log_4 (x - 4)$

j $\log_3 x + \log_3 25x = 8$

l $\log (x + 12) - \log (x + 5) = \log x$

n $\log (x + 5) + \log (x - 2) = \log 8$

11 Solve for y in each of the following equations:

a $\ln (\ln e^{-y}) = \ln 3$

b $e^{-\ln y} = 3$

12 Solve the following equations, rounding your answers to two decimal places.

a $(4.5)^{\ln x} = 10^{0.5}$

b $\ln x = -\sqrt[3]{x + 8}$

13 Use a graphing calculator to solve the following equations. Round your answers to the two decimal places where necessary.

a $(\log_3 x)^2 - \log_3 (x^4) + 4 = 0$

b $3 \log^2 x - 2 \log x = 8$

c $\ln x = -\sqrt[3]{x + 9}$

